

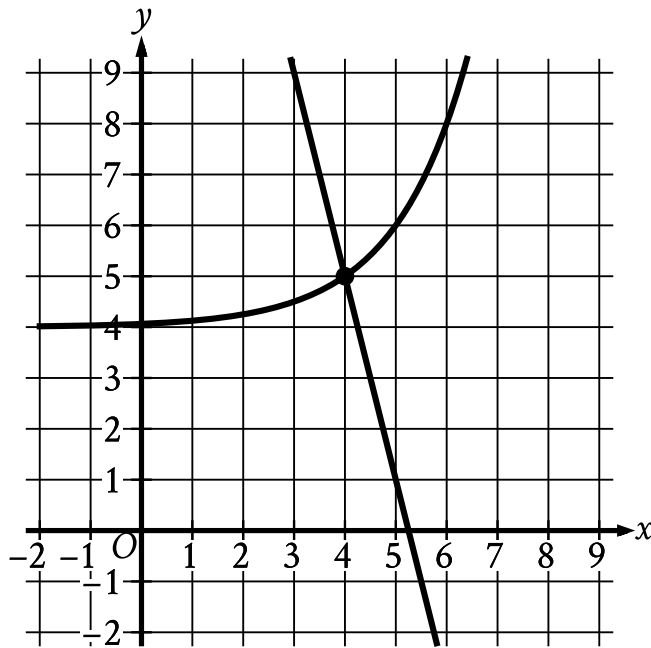
● EASY

● ADVANCED MATH

Advanced Math

10 questions · ~15 min

12



- For the curve in the system:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2, the curve trends up gradually to the approximate point (0 comma 4.063).
 - In quadrant 1:
 - The curve trends up gradually to point (4 comma 5).
 - The curve then trends up sharply.
 - The curve passes through the following points:
 - (3 comma nine halves)
 - (4 comma 5)
 - (5 comma 6)
- For the line in the system:
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - (3 comma 9)
 - (4 comma 5)

The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution x, y to this system?

- A. 0,0
- B. 0,4
- C. 4,5
- D. 5,0

Q2

ADVANCED MATH

Which expression is equivalent to $m^4q^4z^{-1}mq^5z^3$, where m , q , and z are positive?

- A. $m^4q^{20}z^{-3}$
- B. $m^5q^9z^2$
- C. $m^6q^8z^{-1}$
- D. $m^{20}q^{12}z^{-2}$

The function f is defined by $f(x) = 6 + \sqrt{x}$. What is the value of $f(36)$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

ADVANCED MATH

$$x^2 = 64$$

Which of the following values of x satisfies the given equation?

- A. -8
- B. 4
- C. 32
- D. 128

Q5

ADVANCED MATH

Which expression is equivalent to $9x^2 + 7x^2 + 9x$?

- A. $63x^4 + 9x$
- B. $9x^2 + 16x$
- C. $25x^5$
- D. $16x^2 + 9x$

The function f is defined by $fx = \frac{1}{6x}$. What is the value of fx when $x = 3$?

A. $\frac{1}{3}$

B. $\frac{1}{6}$

C. $\frac{1}{9}$

D. $\frac{1}{18}$

Q7

GRID-IN

ADVANCED MATH

$$x - 5 = 10$$

What is one possible solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

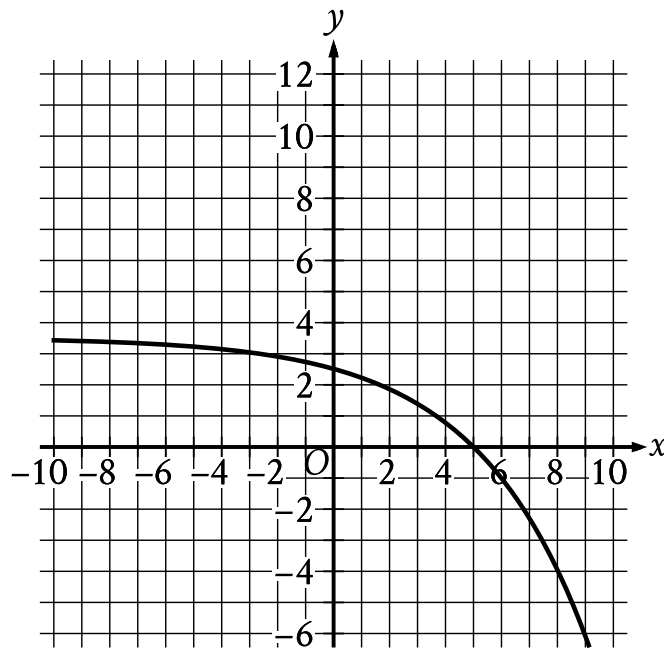
Which expression is equivalent to $19x^2 - 7$?

A. $19x^2 - 133$

B. $19x^2 - 26$

C. $19x^2 - 7$

D. $19x^2 + 12$



- Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1 to quadrant 4.
 - In quadrant 2, the curve trends down gradually to the approximate point (0 comma 2.5).
 - In quadrant 1, the curve trends down sharply to point (5 comma 0).
 - In quadrant 4, the curve trends down sharply.
- The curve passes through the following points:
 - approximately (0 comma 2.5)
 - approximately (4 comma 0.8)

What is the x -intercept of the graph shown?

A. -5,0

B. 5,0

C. $-2, 0$

D. $2, 0$

Q10

ADVANCED MATH

$$6x - 9y > 12$$

Which of the following inequalities is equivalent to the inequality above?

A. $x - y > 2$

B. $2x - 3y > 4$

C. $3x - 2y > 4$

D. $3y - 2x > 2$